

Junseo Lee – Curriculum Vitae

CONTACT INFORMATION

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EDUCATION

Harvard University, Kenneth C. Griffin Graduate School of Arts and Sciences (GSAS) Cambridge, MA
Ph.D. in Quantum Science and Engineering Sep. 2026 – Present

- Advisors: [Prof. Anurag Anshu](#) and [Prof. Sitan Chen](#) (Computer Science); [Prof. Jordan Cotler](#) (Physics)
- Supported by the Kwanjeong Ph.D. Fellowship

Yonsei University Seoul, Korea
B.S. in Electrical and Electronic Engineering Mar. 2019 – Feb. 2023

- Supported by the Hyundai Motor CMK Science and Technology Scholarship

Chungnam Science High School Gongju, Korea
Concentration in Mathematics; Early Honors Graduate Mar. 2017 – Dec. 2018

RESEARCH POSITIONS

NSF AI Institute for Artificial Intelligence and Fundamental Interactions Cambridge, MA
Junior Investigator (Nominator/Advisor: Prof. Jordan Cotler) May. 2026 – Present

Institute of Computer Technology, Seoul National University Seoul, Korea
Research Fellow (Host: Dr. Kabgyun Jeong) Apr. 2026 – Aug. 2026

Research Institute of Mathematics, Seoul National University Seoul, Korea
Research Affiliate (Host: Dr. Kabgyun Jeong) Jan. 2023 – Mar. 2026
Undergraduate Research Assistant (Advisor: Dr. Kabgyun Jeong) Mar. 2020 – Dec. 2022

Republic of Korea Army (Mandatory Military Service) Seoul, Korea
Research Scientist (Technical Research Personnel) Mar. 2023 – Mar. 2026

PUBLICATIONS ([Google Scholar](#), [arXiv](#))

*Equal contribution. †Authors listed alphabetically.

- (19) Francisco Escudero Gutiérrez, **Junseo Lee**[†], Sebastian Zur,
“Hamiltonian locality testing and certification do not achieve the Heisenberg limit,”
Manuscript (2026). [arXiv:2610.03205](#).
- (18) **Junseo Lee**[†], Chirag Wadhwa,
“Optimal purity estimation with incoherent measurements,”
Manuscript (2026). [arXiv:2609.40248](#).
- (17) Myeongjin Shin, **Junseo Lee**, Iman Marvian, Yu Tong,
“Emergent prethermal symmetries for scalable Hamiltonian learning,”
Manuscript (2026). [arXiv:2609.27181](#).
- (16) Myeongjin Shin, **Junseo Lee**, Changhun Oh,
“Heisenberg-limited Hamiltonian learning without short-time control,”
[PRX Quantum](#) (2026). DOI: 10.1103/kjvf-jgqj.
Long talk at the [26th Asian Quantum Information Science Conference \(AQIS 2026\)](#).

- (15) Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, **Junseo Lee**[†], Aadil Oufkir, Cambyse Rouzé, Myeongjin Shin,
“Certifying and learning local quantum Hamiltonians,”
Manuscript (2026). arXiv:2603.29809.
Contributed talk at the [21st Conference on the Theory of Quantum Computation, Communication and Cryptography \(TQC 2026\)](#).
- (14) **Junseo Lee**[†], Myeongjin Shin,
“Optimal certification of constant-local Hamiltonians,”
Manuscript (2025). arXiv:2512.09778.
- (13) Marco Fanizza, Vishnu Iyer, **Junseo Lee**[†], Antonio Anna Mele, Francesco Anna Mele,
“Efficient learning of bosonic Gaussian unitaries,”
Manuscript (2025). arXiv:2510.05531.
Contributed talk at the [29th Annual Conference on Quantum Information Processing \(QIP 2026\)](#).
- (12) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, Kabgyun Jeong,
“Bounding quantum uncommon information with quantum neural estimators,”
[Quantum Science and Technology 11, 015001 \(2026\)](#). DOI: 10.1088/2058-9565/ae18f4.
- (11) Kartik Anand, Kabgyun Jeong, **Junseo Lee**[†],
“Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy,”
Manuscript (2025). arXiv:2506.19792.
- (10) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, Kabgyun Jeong,
“Resource-efficient algorithm for estimating the trace of quantum state powers,”
[Quantum 9, 1832 \(2025\)](#). DOI: 10.22331/q-2025-08-27-1832.
- (9) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, Kabgyun Jeong,
“Mutual information maximizing quantum generative adversarial networks,”
[Scientific Reports 15, 32835 \(2025\)](#). DOI: 10.1038/s41598-025-18476-y.
- (8) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, Kabgyun Jeong,
“Disentanglement provides a unified estimation for quantum entropies and distance measures,”
[Physical Review A 110, 062418 \(2024\)](#). DOI: 10.1103/PhysRevA.110.062418.
- (7) Myeongjin Shin, **Junseo Lee**, Kabgyun Jeong,
“Estimating quantum mutual information through a quantum neural network,”
[Quantum Information Processing 23, 57 \(2024\)](#). DOI: 10.1007/s11128-023-04253-1.
- (6) **Junseo Lee**, Kabgyun Jeong,
“Quantum Rényi entropy functionals for bosonic gaussian systems,”
[Physics Letters A 490, 129183 \(2023\)](#). DOI: 10.1016/j.physleta.2023.129183
- (5) **Junseo Lee**, Hyeonjun Yeo, Kabgyun Jeong,
“Weighted p -Rényi entropy power inequality: Information theory to quantum Shannon theory,”
[International Journal of Theoretical Physics 62, 253 \(2023\)](#). DOI: 10.1007/s10773-023-05512-8
- (4) **Junseo Lee**, Kabgyun Jeong,
“High-dimensional private quantum channels and regular polytopes,”
[Communications in Physics 31, 189 \(2021\)](#). DOI: 10.15625/0868-3166/15762
- (3) Kabgyun Jeong, **Junseo Lee**, et al.,
“Single qubit private quantum channels and 3-dimensional regular polyhedra,”
[New Physics: Sae Mulli 68, 232 \(2018\)](#). DOI: 10.3938/NPSM.68.232
Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

- (2) **Junseo Lee**,
“Assessing Quantum Integer Factorization Performance with Shor’s Algorithm,”
In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*,

CRC Press (2024). DOI: 10.1201/9781003475286

Patents

- (1) Kabgyun Jeong, Myeongjin Shin, **Junseo Lee**,
“Method for estimating quantum mutual information through a quantum neural network,”
Korea Patent Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

SELECTED HONORS AND AWARDS

Funding, Scholarships and Fellowships

- Ph.D. Fellowship, *Kwanjeong Foundation*, 2026–2030
- QSE Professional Development Fellowship, *Harvard University*, 2026
- Academic Travel Grant, *Hyundai Motor CMK Foundation*, 2022
- Science and Technology Scholarship (Full Tuition and Stipend), *Hyundai Motor CMK Foundation*, 2021–2022

Academic Excellence

- High Honors, *Yonsei University*, 2022
- Honors, *Yonsei University*, 2020–2021
- Science Scholarship, *JH Foundation*, 2017
- Academic Excellence Scholarship, *Asan Future Scholarship Foundation*, 2016

Additional Honors and Awards

- Selected Paper Award, Finance and Economics Contest, *DB Cultural Foundation*, 2022
- Best Tutor Award, *Yonsei University*, 2021–2022
- Third Prize, Undergraduate Research Exhibition, *Korean Physical Society*, 2021
- Bronze Award, Humantech Paper Award, *Samsung Electronics*, 2018
- Best Translator Award, *NAVER Connect Foundation & Khan Academy*, 2018
- Pre-Undergraduate Research Fellowship, *Korea Advanced Institute of Science and Technology*, 2017–2018
- Honorable Mention (National) & Gold Award (Regional), Korean Olympiad in Informatics (KOI), 2016

PROFESSIONAL ACTIVITIES

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Portfolio:** Nature Communications, npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik, Mathematical Methods in the Applied Sciences, Statistics and Probability Letters, Algorithmica

Conference Reviewer

- ACM-SIAM Symposium on Discrete Algorithms (SODA), 2027
- IEEE Symposium on Foundations of Computer Science (FOCS), 2026
- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- Organizer, [Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems](#), 2026
- Co-organizer, [Quantum Information Theory Seminar](#), *Seoul National University*, 2024–2025

- Co-organizer, [Quantum AI Hackathon](#), *Kakao Enterprise & Jeonju University*, 2025

Committee Service

- Poster Session Committee Member, National Undergraduate Quantum Conference, *Seoul National University*, 2026
- Selection Committee Member, [Quantum Internship Program](#), *National Information Society Agency & Korea Quantum Industry Center*, 2024 – 2025

Outreach and Community Resources

- Creator and Maintainer, [Quantum Learning Theory Zoo](#), 2025–present
- Facilitator (Mentor), Korea Scholar's Conference for Youth (KSCY), *Yonsei University*, 2019

TEACHING

*Teaching materials available at: harris-junseo-lee.github.io. †University–Industry Research Internship.

- Organizer and Instructor, [Quantum Learning Theory for Bosonic and Fermionic Systems](#), *Quantum Information Science Club Association*, Winter 2026*
- Instructor, [Quantum Complexity Reading Group](#), *Quantum Information Science Club Association*, Fall 2025*
- Instructor, [Quantum Learning and Complexity Theory](#), *Quantum Information Science Club Association*, Summer 2025*
- (External) Instructor†, AAA559: College of Informatics Internship (II) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, AAA558: College of Informatics Internship (I) (Graduate Course), *Korea University*, Fall 2025
- (External) Instructor†, SW4343: Software Field Placement (I), *Korea Aerospace University*, Fall 2024
- Teaching Assistant, YCS1009: Change the World through Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, YCS1002: Software Programming, *Yonsei University*, Fall 2022
- Teaching Assistant, EEE1108: Engineering Information Processing, *Yonsei University*, Fall 2021
- Course Tutor, MAT2016: Engineering Mathematics (III), *Yonsei University*, Spring 2022 (Best Tutor Award)
- Course Tutor, MAT1012: Engineering Mathematics (II), *Yonsei University*, Fall 2021 (Best Tutor Award)

RESEARCH VISITS

- Harvard University (supported by Prof. Anurag Anshu and the Harvard Quantum Initiative), Mar. 2026

SELECTED TALKS

*Online talk.

Research Talks

“Heisenberg-limited Hamiltonian learning without short-time control”

- Invited talk, *Technische Universität Hamburg (Hamburg University of Technology)*, Jun. 2026*
- Long contributed talk, 26th Asian Quantum Information Science Conference (**AQIS 2026**), *Korea Advanced Institute of Science and Technology (KAIST)*, Aug. 2026

“Certifying and learning local quantum Hamiltonians”

- Invited talk, Quantum Meets (Online Seminar Series), Jun. 2026*
- Invited talk, 7th QNRC Joint Workshop, *Korea Institute of Science and Technology Information (KISTI)*, May 2026
- Contributed talk, 21st Conference on the Theory of Quantum Computation, Communication and Cryptography (**TQC 2026**), *Delta Hotels Sherbrooke Conference Centre (Québec, Canada)*, Sep. 2026

“Optimal certification of constant-local Hamiltonians”

- Invited talk, Quantum Software Lab Seminar, *University of Edinburgh*, Mar. 2026*
- Contributed talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026

“Efficient learning of bosonic Gaussian unitaries”

- Invited talk, Annual Meeting of the Quantum Information Society of Korea (QISK 2026), *Seoul SETEC Convention Center*, Feb. 2026
- Invited talk, N³etFraST Workshop, *Korea Institute of Science and Technology Information (KISTI)*, Nov. 2025
- Invited talk, Quantum Data Science & AI Lab Seminar, *Yonsei University*, Nov. 2025
- Contributed talk, 29th Annual Conference on Quantum Information Processing (**QIP 2026**), *University of Latvia*, Jan. 2026 (presented under the title “Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators”)

“Resource-efficient algorithm for estimating the trace of quantum state powers”

- Invited talk, *Electronics & Telecommunications Research Institute (ETRI)*, Dec. 2024
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Dec. 2024*
- Invited talk, IBM-Yonsei Qiskit Fall Fest, *Yonsei University*, Nov. 2024*
- Invited talk, KISTI-KU-SNU Joint Workshop, *Seoul Biohub*, Oct. 2024
- Contributed talk, Annual Meeting of the Korean Mathematical Society (KMS), *Sungkyunkwan University*, Oct. 2024
- Poster, 28th Annual Conference on Quantum Information Processing (QIP 2025), *Raleigh Convention Center (NC, United States)*, Feb. 2025

“Mutual information maximizing quantum generative adversarial network”

- Invited talk, Triangle Quantum Computing Seminar, *NC State University Quantum Initiative*, Nov. 2023*

“Estimating quantum mutual information through a quantum neural network”

- Invited talk, CS Katha Barta, *National Institute of Science Education and Research (Bhubaneswar, India)*, Aug. 2023*

“Quantum Rényi entropy functionals for bosonic Gaussian systems”

- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

“High-dimensional private quantum channels and regular polytopes”

- Invited talk, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- Invited talk, Quantum Information Theory Seminar, *Seoul National University*, Aug. 2021*
- Contributed talk, Winter Meeting of the Optical Society of Korea (OSK), *Daejeon Convention Center*, Feb. 2022
- Contributed talk, Fall Meeting of the Korean Physical Society (KPS), Oct. 2021*
- Poster, 25th Annual Conference on Quantum Information Processing (QIP 2022), *Caltech*, Mar. 2022

Tutorials and Public Lectures

“Getting Started in Quantum Information Science”

- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Apr. 2026
- Invited talk, Student Outreach Lecture, *Shinil High School (Seoul, Korea)*, Aug. 2024

“Learning theory in ∞ -dimensional quantum systems”

- Invited talk, Team QST Summer Workshop, *Seoul National University*, Aug. 2025

“Introduction to quantum machine learning”

- Invited talk, Healthcare & Research Team Seminar, *Amazon Web Services (AWS Seoul, Korea)*, Mar. 2025

“Quantum machine learning models for drug library generation”

- Invited talk, Quantum Computing and Monte Carlo Workshop, *Yonsei University*, Aug. 2024

“QMA $\stackrel{?}{=}$ NP: The NLTS theorem and the quantum PCP conjecture”

- Invited talk, Center for Quantum Network’s Channel Capacity Summer Workshop, *Seoul National University*, Jul. 2024

“Minimal data may be sufficient for quantum artificial intelligence”

- Invited talk, Department of Mathematical Sciences Seminar, *Seoul National University*, Jun. 2023*